Dealing with Austrian Values in Crisis data for Wave 1, Wave 2 and the panel data

Sampling design, weighting strategies and further index variables (such as ISCO 08) and scales (e.g., values, personality)

Alexander Seymer, Martin Ulrich, Wolfgang Aschauer & Markus Kreuzberger[1] in cooperation with

Manfred Herzog[2] & Otto Bodi Fernandez[3]

1. Introduction

The following document describes the sampling design and the weighting strategy for the first and the second wave of the Austrian Values in Crisis (VIC Austria) survey. Additionally further index variables (such as the occupational classification) and value and personality scales which are part of the data are further described. The sample includes respondents from wave 1 (W1, n=2018), from wave 2 (W2, n=2082) and respondents participating in both waves – the panel data (PD).

The report consists of five different parts. The section on sampling (part 2) describes the main features of the online-access panel study. The document further describes the computation of three different weights corresponding to the respective samples (W1, W2, PD). First the target population will be characterized. Next, the applied sampling strategy is explained in detail to set the stage for the discussion of coverage and sampling errors and the corrections by post-stratification weights.

After reporting the key figures on our adjustment weights some remarks how to use the ISCO variables for wave 2 at the end of the dataset are given (section 4). Finally the report explains with more details how the main value and personality scores were computed and how to ideally deal with these variables.

Besides those scales bot studies (VIC wave 1 and VIC wave 2) integrate a lot of item batteries referring to attitudes towards politics and perceptions on public health, as well as the own exposure to the pandemic and perceptions of social and economic consequences for Austrian society. The second wave (fieldwork period 25th of March till the 12th of April) has many repeated modules but additionally includes specific questions about attitudes towards public restrictions, towards social distancing or Austrian perceptions on social integration.[4]

2. What population does VIC Austria represent?

The survey was conducted by the Marketagent.com online reSEARCH GmbH (Marketagent)[5] based on a consumer online panel with 130,500 participants in Austria. The online panel integrates persons who are at least 14 years or older. All educational groups, professions and Austrian regions are well represented although people in older age groups and immigrants are underrepresented. The participants speak (one of) the official language(s) of the country (regardless of nationality and

[1] University of Salzburg, Department of Sociology and Cultural Science

[2] AUSSDA - The Austrian Social Science Data Archive, University of Vienna

[3] AUSSDA - The Austrian Social Science Data Archive, University of Graz

[4] Please see both questionnaires which are also publicly released to get an overview of the survey.

[5] Marketagent.com. (2021). *Panel Book* [Brochure]. Baden, Austria: Author. Retrieved July 26, 2021, from https://b2b.marketagent.com/media/itkf4i0o/marketagent-panel-book-2021.pdf

citizenship) and do not live either abroad or in institutions such as prisons and hospitals during the entire fieldwork period.

Table 1 illustrates major deviations of the Online-Panel compared to Austrian statistics. Compared to the target population the Marketagent sample has more women and people between 20 and 39 years. Additionally, samples drawn from the panel recruiting people 60 years and more will rely on a smaller pool than younger age groups. Similar deviations from the target population appear regarding marital status, household size and number of children in the household. Married and widowed people are underrepresented as well as people in single person households and households without children. Obviously, these deviations relate to the age distortion as older people are more likely to be widowed and to live in a single person household without children.

Table 1: Comparison sampling frame 2020 and 2021 and target population 2020 and 2021

	Characteristic	**Marketagent Panel 2020[1]**	**Marketagent Panel 2021[2]**	**Statistics Austria 2020[3]**	**Statistic Austria 2021[3]**
Female	14-19 years	2.8	1.5	3.3	3.3
	20-29 years	15.7	14.8	7.1	6.9
	30-39 years	16.8	17.6	7.8	7.8
	40-49 years	9.9	10.7	7.6	7.7
	50-59 years	7.1	7.4	9.1	9.0
	60+ years	3.8	4.5	16.2	16.4
	Sum	56.2	56.6	51.2	51.1
Male	14-19 years	2.6	1.3	3.5	3.5
	20-29 years	12.0	11.4	7.5	7.3
	30-39 years	10.6	11.4	8.0	8.0
	40-49 years	7.6	7.8	7.7	7.6
	50-59 years	6.0	6.1	9.1	9.0
	60+ years	5.1	5.4	13.1	13.4
	Sum	43.8	43.4	48.8	48.9
Marital status	Single	58.3	54.4	30.7	n/a
	Married/partnership	30.6	35.5	45.8	n/a
	Divorced	9.9	9.0	13.7	n/a
	Widowed	1.2	1.1	9.8	n/a
Household size	Single person	20.3	20.4	37.8	n/a
	Two persons	33.2	33.3	30.4	n/a
	Three and more	46.5	46.3	31.9	n/a
Children per household	None	56.5	57.0	77.4	n/a
	One	35.2	34.9	11.0	n/a
	Two or more	8.3	8.1	11.7	n/a

Note: [1] Marketagent 2020; [2] Marketagent 2021; [3] Statistics Austria relies on different databases: Gender and Age (Statistics Austria 2021a); Marital status, Household size and Children in Household (Statistics Austria 2020)

3. The weighting procedure in the Austrian VIC Survey

The VIC Austria is part of a cross-national survey and as most modern population surveys, complex sampling strategies are applied to improve survey quality. Considering these sampling procedures is particularly important for cross-national surveys like the VIC study in order to mirror country characteristics adequately.[6] A commonly applied method is the combination of quota sampling and poststratification weighting based on register data to correct for the different sampling probabilities.[7] [8] [9] [10]

In the online panel a stratified quota sampling strategy was applied for gender, age, region and education. The distribution of the unweighted final sample of Wave 1 is presented in Table 2, for Wave 2 in and for the panel data in Error: Reference source not found. As a result of the quotas, the sample represents the target population already rather well. Still the sampling probabilities for the different groups vary due to the mismatch between sampling frame and target population. This will especially impose a bias for the age group 70+ and lower education groups in Wave 2.

[6] Kish, L. (1994). Multipopulation Survey Designs: Five Types with Seven Shared Aspects. International Statistical Review / Revue Internationale de Statistique, 62(2), 167–186. JSTOR. https://doi.org/10.2307/1403507

[7] DeBell, M., & Krosnick, J. A. (2009). Computing Weights for American National Election Study Survey Data. Retrieved July 27, 2021, from https://electionstudies.org/wp-content/uploads/2018/04/nes012427.pdf

[8] Lynn, P., & Anghelescu, G. (2018). Europoean Social Survey Round 8 Weighting Strategy. Retrieved July 27, 2021, from https://www.europeansocialsurvey.org/sites/default/files/2023-06/ESS8_weighting_strategy_0.pdf

[9] Statistics Austria (2021a). STATcube – statistical database of Statistics Austria. Population at the beginning of the year from 2002 onwards (uniform territorial status 2020; data are continuously imported into a population statistics database system), based on the Central Register of Residents. Retrieved Juli 26, 2021, from https://statcube.at/statistik.at/ext/statcube/jsf/terms.xhtml

[10] Statistics Austria (2021b). STATcube – statistical database of Statistics Austria. Register-based Labour Market Statistics - Persons - Time Series since 2011. Retrieved Juli 27, 2021, from http://www.statistik.at/web_en/statistics/PeopleSociety/education/educational_attainment/index.html

Table 2: Comparison unweighted sample and target population for Wave 1

	Characteristic	Unweighted Sample	Statistics Austria 2020
Female[1]	14-19 years	3.8	3.3
	20-29 years	7.8	7.1
	30-39 years	8.1	7.8
	40-49 years	8.7	7.6
	50-59 years	9.3	9.1
	60-69 years	7.1	6.8
	70+ years	5.9	9.4
	Sum	50.8	51.2
Male[1]	14-19 years	3.7	3.5
	20-29 years	8.2	7.5
	30-39 years	7.9	8.0
	40-49 years	8.5	7.7
	50-59 years	9.4	9.1
	60-69 years	6.5	6.3
	70+ years	5.0	6.8
	Sum	49.2	48.8
Region[1]	Vorarlberg	4.4	4.5
	Tyrol	8.6	8.5
	Salzburg	6.2	6.3
	Styria	14.3	14.0
	Carinthia	6.5	6.3
	Upper Austria	16.9	16.7
	Lower Austria	18.9	18.9
	Vienna	20.5	21.5
	Burgenland	3.7	3.3
Education[2]	Compulsory education	17.2	25.1
	Apprenticeship	38.7	31.3
	Vocational middle school	13.1	13.9
	General or vocational secondary school	19.7	15.3
	University	11.4	14.4

Note: [1] Statistics Austria 2021a; [2] Statistics Austria 2021b.

Table 3: Comparison unweighted sample and target population for Wave 2

	Characteristic	Unweighted Sample	Statistics Austria 2021
Female[1]	14-19 years	3.1	3.3
	20-29 years	8.2	6.9
	30-39 years	8.0	7.8
	40-49 years	8.4	7.6
	50-59 years	9.3	9.0
	60-69 years	7.4	7.0
	70+ years	6.0	9.4
	Sum	50.4	51.1
Male[1]	14-19 years	2.6	3.5
	20-29 years	8.4	7.3
	30-39 years	7.8	8.0
	40-49 years	8.4	7.6
	50-59 years	9.6	9.0
	60-69 years	7.7	6.5
	70+ years	5.1	6.8
	Sum	49.6	48.9
Region[1]	Vorarlberg	4.0	4.5
	Tyrol	8.1	8.5
	Salzburg	6.3	6.3
	Styria	14.3	14.0
	Carinthia	6.3	6.3
	Upper Austria	17.5	16.7
	Lower Austria	18.9	18.9
	Vienna	20.9	21.5
	Burgenland	3.6	3.3
Education[2]	Compulsory education	14.0	25.1
	Apprenticeship	45.8	31.3
	Vocational middle school	10.8	13.9
	General or vocational secondary school	18.7	15.3
	University	10.8	14.4

Note: [1] Statistics Austria 2021a; [2] Statistics Austria 2021b.

Table 4: Comparison unweighted sample and target population for panel data

	Characteristic	Unweighted Sample	Statistics Austria 2021
Female[1]	14-19 years	1.3	3.3
	20-29 years	4.3	6.9
	30-39 years	7.6	7.8
	40-49 years	8.6	7.6
	50-59 years	10.7	9.0
	60-69 years	9.5	7.0
	70+ years	8.1	9.4
	Sum	50.2	51.1
Male[1]	14-19 years	0.9	3.5
	20-29 years	5.7	7.3
	30-39 years	6.3	8.0
	40-49 years	9.4	7.6
	50-59 years	11.3	9.0
	60-69 years	9.1	6.5
	70+ years	7.1	6.8
	Sum	49.8	48.9
Region[1]	Vorarlberg	4.3	4.5
	Tyrol	8.3	8.5
	Salzburg	6.1	6.3
	Styria	15.1	14.0
	Carinthia	5.7	6.3
	Upper Austria	16.4	16.7
	Lower Austria	19.9	18.9
	Vienna	20.4	21.5
	Burgenland	3.8	3.3
Education[2]	Compulsory education	14.7	25.1
	Apprenticeship	43.5	31.3
	Vocational middle school	11.9	13.9
	General or vocational secondary school	18.2	15.3
	University	11.8	14.4

Note: [1] Statistics Austria 2021a; [2] Statistics Austria 2021b.

To correct for deviations between sample and target population, survey sample weights were calculated by raking aligning the data to population margins. The reference population margins are gender, age and education and region separately (see Table 2, and Error: Reference source not found). The R package anesrake by Josh Pasek (2018) was used to estimate the weights.[11]

The calculated weight variables for the different samples have a standard deviation of 0.339 (W1), 0.460 (W2), 0.521 (PD); a minimum of 0.461 (W1), 0.494 (W2), 0.532 (PD); and a maximum of 2.666 (W1), 3.498 (W2), 4.940 (PD). For the W1 weight no weight exceeded values of 0.3 on the lower end and 3 on the upper. For the W2 weight 5 cases had weights larger than 3. The 6 cases scale up to 16.6 cases when the weight is applied. Similarly, the panel data weights have 11 cases with weights greater than 3. These 11 scale up to 45.0 cases in the weighted panel data sample. The W2 and the

[11] Pasek, J. (2018). anesrake R package version 0.8. https://cran.r-project.org/web/packages/anesrake/anesrake.pdf

PD weighting scores are considered moderately skewed and no adjustment was made to the weights to account for outliers.[12]

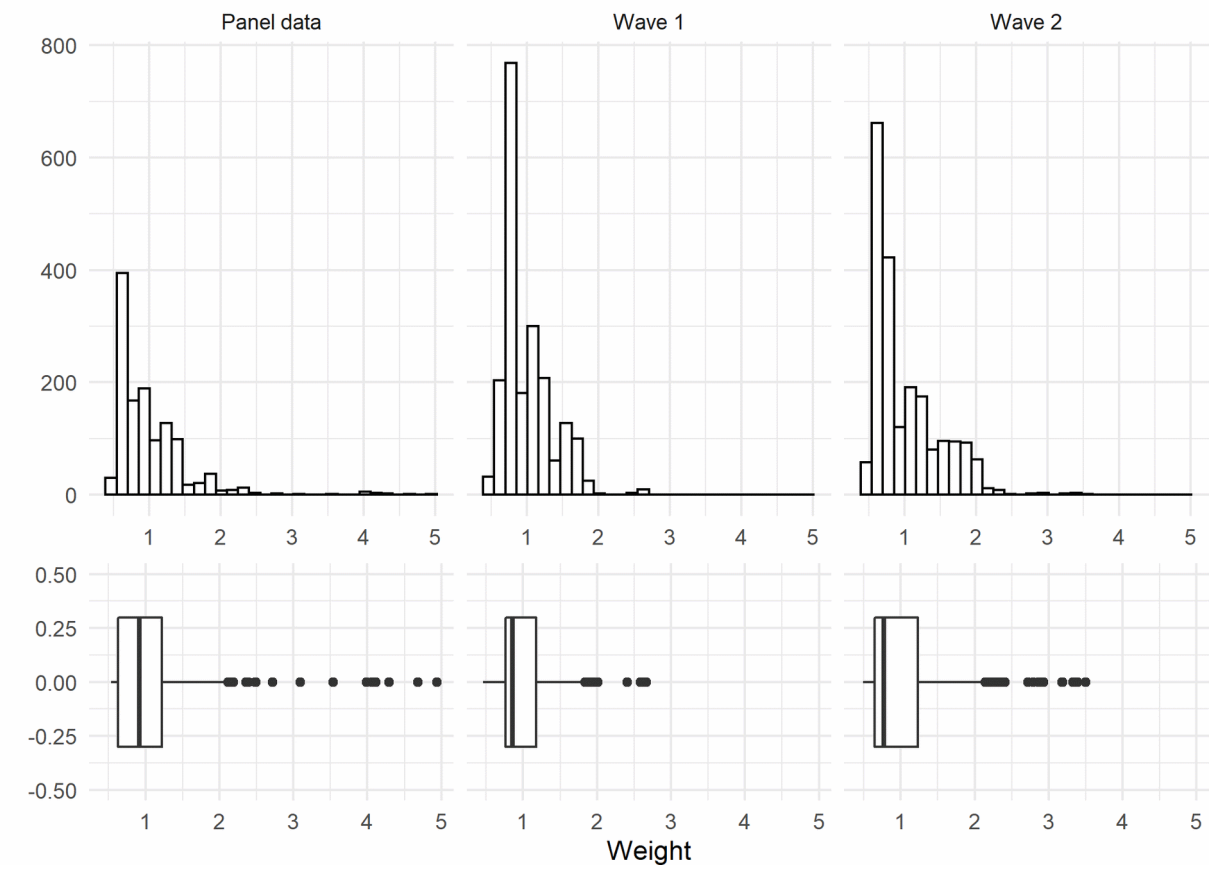

Figure 1: Distribution of the weights

Figure 1 presents the distributions of the weights. Wave 1 performs best with the smallest weights and the panel data has the largest dispersion. Wave 1 and 2 have been sampled according to quotas with respect to gender, age, education and region. Accordingly, the samples represent the population already well. For the panel data, the selection is based on the willingness of respondents to Wave 1 to participate in Wave 2. Hence, the sample deviates from the population more significantly leading to larger weights.

Table 5 compares the unweighted (raw) and the weighted (adjusted) data for the four variables used in the weight estimation. The comparison of weighted to unweighted distributions emphasizes the small size in adjustments made by the weights. Only with respect to the distribution of lower education groups the weights shift the unweighted data to a larger extend. More precisely, respondents with compulsory education are underrepresented in the sample. As older age groups have more often only compulsory education compared to younger cohorts, the older age groups are slightly underrepresented as well.

Table 6 to 8 compares the weighted data in a three-way table of gender, age and education against the unweighted data. The largest deviations between weighted and unweighted distribution can be observed in the columns Compulsory school and Apprenticeship. The deviations on the other groups

[12] Little, R. J. A., Lewitzky, S., Heeringa, S., Lepkowski, J., & Kessler, R. C. (1997). Assessment of Weighting Methodology for the National Comorbidity Survey. American Journal of Epidemiology, 146(5), 439–449. https://doi.org/10.1093/oxfordjournals.aje.a009297

are minor. In general, the adjustments made by the weights are considerable small and should be unproblematic with respect to potential adjustment errors.

Table 5: Comparison of unweighted (raw) and weighted (adjusted) data.

	Characteristic	Wave 1		Wave 2		Panel data	
		raw	adjusted	raw	adjusted	raw	adjusted
Gender[1]	Female	50.8	51.2	50.4	51.1	50.2	51.1
	Male	49.2	48.8	49.6	48.9	49.8	48.9
Age[1]	14-19 years	7.5	6.8	5.8	6.7	2.2	6.7
	20-29 years	16.0	14.5	16.7	14.3	10.0	14.3
	30-39 years	16.0	15.8	15.8	15.9	13.9	15.9
	40-49 years	17.3	15.4	16.8	15.3	17.9	15.3
	50-59 years	18.7	18.1	18.9	18.1	22.0	18.1
	60-69 years	13.5	13.2	15.0	13.6	18.6	13.6
	70+ years	11.0	16.1	11.1	16.2	15.4	16.2
Region[1]	Vorarlberg	4.4	4.5	4.0	4.5	4.3	4.5
	Tyrol	8.6	8.5	8.1	8.5	8.3	8.5
	Salzburg	6.2	6.3	6.3	6.3	6.1	6.3
	Styria	14.3	14.0	14.3	14.0	15.1	14.0
	Carinthia	6.5	6.3	6.3	6.3	5.7	6.3
	Upper Austria	16.9	16.7	17.5	16.7	16.4	16.7
	Lower Austria	18.9	18.9	18.9	18.9	19.9	18.9
	Vienna	20.5	21.5	20.9	21.5	20.4	21.5
	Burgenland	3.7	3.3	3.6	3.3	3.8	3.3
Education[2]	Compulsory education	17.2	25.1	14.0	25.1	14.7	25.1
	Apprenticeship	38.7	31.3	45.8	31.3	43.5	31.3
	Vocational middle school	13.1	13.9	10.8	13.9	11.9	13.9
	General or vocational secondary school	19.7	15.3	18.7	15.3	18.2	15.3
	University	11.4	14.4	10.8	14.4	11.8	14.4

Note: [1] Statistics Austria 2021a; [2] Statistics Austria 2021b, 17 cases matching the category "other degree" have been excluded.

Table 6: Three-way table with shares for gender, age and education comparing raw (unweighted) and adjusted (weighted) data for Wave 1

		Highest completed training (education)											
		Compulsory school		**Apprenticeship**		**Vocational middle school**		**General or vocational secondary school**		**University**		***Total***	
Gender	**Age Groups**	**raw**	**adjusted**	**raw**	**adjusted**	**raw**	**adjusted**	**raw**	**adjusted**	**raw**	**adjusted**	***raw***	***adjusted***
Female	**< 20 years**	2.5	2.8	0.3	0.1	0.1	0.0	0.9	0.4	0.0	0.0	*3.8*	*3.5*
	20-29 years	0.7	1.0	3.9	3.0	0.6	0.6	2.0	1.4	0.7	0.8	*7.8*	*6.9*
	30-39 years	1.4	2.3	4.0	3.1	1.0	1.0	0.9	0.6	0.7	0.8	*8.1*	*7.9*
	40-49 years	2.2	3.0	2.7	1.9	1.4	1.2	1.1	0.7	1.3	1.2	*8.7*	*8.1*
	50-59 years	1.1	1.8	2.9	2.3	2.0	2.0	1.9	1.4	1.2	1.4	*9.3*	*9.1*
	60-69 years	1.0	1.6	3.2	2.6	2.2	2.3	0.3	0.2	0.3	0.3	*7.1*	*7.1*
	70+ years	0.4	1.1	1.4	1.8	1.1	1.8	1.9	2.2	1.0	1.8	*5.9*	*8.7*
Male	**< 20 years**	1.9	2.2	0.5	0.3	0.1	0.1	1.1	0.6	0.0	0.0	*3.7*	*3.2*
	20-29 years	1.4	2.2	2.6	2.0	0.5	0.5	2.9	2.1	0.6	0.7	*8.2*	*7.7*
	30-39 years	1.6	2.5	3.2	2.5	0.7	0.7	1.2	0.9	1.2	1.3	*7.9*	*8.0*
	40-49 years	1.3	1.8	4.3	3.0	0.7	0.6	1.1	0.7	1.1	1.1	*8.5*	*7.3*
	50-59 years	0.8	1.4	4.8	4.0	0.9	1.0	1.8	1.4	1.1	1.3	*9.4*	*9.1*
	60-69 years	0.4	0.7	3.3	2.7	1.1	1.1	1.1	0.8	0.5	0.6	*6.5*	*6.1*
	70+ years	0.1	0.4	1.5	1.9	0.4	0.7	1.4	1.7	1.4	2.6	*5.0*	*7.4*
Total		*17.1*	*25.1*	*38.7*	*31.4*	*13.1*	*13.8*	*19.7*	*15.4*	*11.4*	*14.4*	*100.0*	*100.0*

Note: 17 cases matching the category "other degree" have been excluded.

Table 7: Three-way table with shares for gender, age and education comparing raw (unweighted) and adjusted (weighted) data for Wave 2

		Highest completed training (education)											
		Compulsory school		**Apprenticeship**		**Vocational middle school**		**General or vocational secondary school**		**University**		***Total***	
Gender	**Age Groups**	**raw**	**adjusted**	**raw**	**adjusted**	**raw**	**adjusted**	**raw**	**adjusted**	**raw**	**adjusted**	***raw***	***adjusted***
Female	**< 20 years**	1.4	2.3	0.0	0.0	0.1	0.1	1.3	0.9	0.1	0.1	*3.1*	*3.5*
	20-29 years	0.7	1.2	2.7	1.5	0.5	0.5	2.8	1.9	1.5	1.6	*8.2*	*6.7*
	30-39 years	0.9	1.8	4.6	3.2	0.8	1.0	1.1	1.0	0.6	0.9	*8.0*	*7.9*
	40-49 years	1.7	2.8	3.8	2.2	1.1	1.2	0.8	0.6	1.0	1.1	*8.4*	*7.9*
	50-59 years	1.4	2.5	3.7	2.4	1.9	2.3	1.3	1.0	1.1	1.4	*9.3*	*9.5*
	60-69 years	1.1	1.9	4.0	2.5	1.6	1.8	0.4	0.3	0.3	0.3	*7.4*	*6.8*
	70+ years	0.3	1.1	2.2	2.2	1.2	2.2	1.4	1.7	0.9	1.7	*6.0*	*8.9*
Male	**< 20 years**	1.3	2.2	0.4	0.2	0.0	0.0	0.8	0.6	0.1	0.1	*2.6*	*3.1*
	20-29 years	1.3	2.2	2.8	1.6	0.4	0.5	2.9	2.2	0.9	1.1	*8.4*	*7.5*
	30-39 years	0.9	1.8	4.5	3.3	0.4	0.6	0.9	0.8	1.0	1.4	*7.8*	*8.0*
	40-49 years	1.2	2.1	4.4	2.7	0.7	0.8	1.1	0.8	1.1	1.3	*8.4*	*7.5*
	50-59 years	0.9	1.6	5.7	3.8	0.7	1.0	1.5	1.3	0.8	1.1	*9.6*	*8.7*
	60-69 years	0.5	1.0	4.4	2.9	1.1	1.4	1.3	1.0	0.3	0.4	*7.7*	*6.7*
	70+ years	0.2	0.6	2.6	2.7	0.4	0.8	1.1	1.4	0.9	1.8	*5.1*	*7.3*
Total		*14.0*	*25.1*	*45.9*	*31.3*	*10.8*	*13.9*	*18.7*	*15.3*	*10.7*	*14.3*	*100.0*	*100.0*

Note: 16 cases matching the category "other degree" have been excluded.

Table 8: Three-way table with shares for gender, age and education comparing raw (unweighted) and adjusted (weighted) data for panel data.

		Highest completed training (education)											
		Compulsory school		Apprenticeship		Vocational middle school		General or vocational secondary school		University		*Total*	
Gender	**Age Groups**	**raw**	**adjusted**	**raw**	**adjusted**	**raw**	**adjusted**	**raw**	**adjusted**	**raw**	**adjusted**	***raw***	***adjusted***
Female	**< 20 years**	0.5	2.1	0.0	0.0	0.1	0.3	0.7	1.2	0.1	0.2	*1.3*	*3.9*
	20-29 years	0.5	1.2	2.1	2.4	0.2	0.5	1.1	1.2	0.4	0.7	*4.3*	*6.1*
	30-39 years	1.1	2.1	4.1	3.4	0.4	0.6	1.1	1.0	0.8	1.1	*7.6*	*8.3*
	40-49 years	2.4	3.1	3.0	1.9	1.0	1.0	1.1	0.7	1.1	1.1	*8.6*	*7.8*
	50-59 years	2.0	2.5	3.2	1.9	2.3	2.3	1.6	1.0	1.6	1.6	*10.7*	*9.4*
	60-69 years	1.3	1.6	5.3	3.0	2.0	1.9	0.5	0.2	0.4	0.4	*9.5*	*7.1*
	70+ years	0.4	0.8	2.7	2.2	1.1	1.6	2.3	1.9	1.6	2.1	*8.1*	*8.7*
Male	**< 20 years**	0.5	2.1	0.1	0.2	0.0	0.0	0.3	0.7	0.0	0.0	*0.9*	*3.0*
	20-29 years	0.9	2.1	1.9	2.1	0.4	0.8	2.2	2.5	0.2	0.5	*5.7*	*8.1*
	30-39 years	1.1	2.1	2.8	2.4	0.6	0.8	0.7	0.7	1.1	1.6	*6.3*	*7.5*
	40-49 years	1.6	2.1	4.3	2.5	0.8	0.8	1.3	0.8	1.4	1.4	*9.4*	*7.6*
	50-59 years	1.3	1.7	5.7	3.5	1.1	1.2	2.1	1.2	1.1	1.1	*11.3*	*8.8*
	60-69 years	0.8	1.0	5.1	3.0	1.2	1.1	1.6	0.9	0.4	0.4	*9.1*	*6.4*
	70+ years	0.3	0.6	3.2	2.7	0.6	0.8	1.6	1.3	1.5	2.1	*7.1*	*7.5*
Total		*14.8*	*25.1*	*43.5*	*31.3*	*11.9*	*13.9*	*18.1*	*15.3*	*11.7*	*14.4*	*100.0*	*100.0*

Note: 12 cases matching the category "other degree" have been excluded.

4. Implementing the International Standard Classification of Occupations (ISCO-08) in VIC II

The second wave also includes the ISCO-08 classification for the respondents and – if available – also for their partners. ISCO-08 is an updated version of the ISCO-88 classification. This update was necessary because the ISCO-88 classification was not able to accurately represent the job market in the face of the rapid development in the information and communication technology sector. Also, the ISCO-08 provides a more detailed picture of the occupational structure in each country. A comprehensive list for the categories and their definitions can be found at the website of the International Labor Organization.[13] [14]

In the second wave of the VIC survey respondents were asked to state the name of their current or past occupation as open-end questions (Q27 and Q28). Additionally, the respondents shared their formal educational level (Q15). The same information was inquired about the respondents' partners (Q33, Q34 and Q25). Using this information, we categorized each of the respondents' open-end answers into the ISCO-08 categories by hand.[15] The basis for assigning respondents to their respective groups was a document from Statistic Austria that gives detailed accounts about the typed of occupations for each (sub)group for the case of Austria.[16]

Missing values are coded with 9999, answers that could not be assigned to any group are coded with 9998. Respondents that are still in training or education are coded as -9999.

The ISCO-08 classification comes in four different types. The broadest typification (single digit) divides the respondents into ten major groups (see table 9). Each additional digit further differentiates them. The second digit adds information about the occupation group (e.g., in group 5: personal service workers vs. sales workers), the third digit gives information about which occupational subgroup the respondent belongs to (e.g. in group 51: cooks vs. waiters and bartenders).[15] We provide three different variables about the ISCO-08 classification in the Value in Crisis dataset: the one-, two- and three-digit classification for both the respondents and their partners.

The higher the number of digits in the ISCO-08 classification, the more cases are codes as 9998. The reason for that is that while the information a respondent gives is sufficient to classify them in the single-digit variable, their answer may not be descriptive or unambiguous enough to do the same in the two- or three-digit variable. This is why the amount of 9998 codes slightly rises from 98 cases in ISCO1 to 109 cases in ISCO2 and 120 cases in ISCO3.

The ISCO-08 classification can also be used to calculate several indices regarding social positions (e.g., ISEI-08, SIOPS, etc.). For this, we recommend using the SPSS-syntax files by Ganzeboom (2010).[17] [18]

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[13] ISCO - International Standard Classification of Occupations (ilo.org) (last accessed 08/18/2021)

[14] Bureau of Statistics, work unit of the Policy Integration Department (ilo.org) (last accessed 08/18/2021)

[15] Züll, C. (2015). Berufscodierung. Retrieved August 18, 2021, from Berufscodierung_Zuell_08102015_1.1.pdf (gesis.org)

[16] Vgl. Statistik Austria Erläuterungen zu ISCED 08 (last accessed 08/18/2021)

[17] HARRY GANZEBOOM'S Tools for occupation status measures for ISCO-68 (last accessed 08/18/2021)

[18] Ganzeboom, H. B. G. (2010). A new international socio-economic index (ISEI) of occupational status for the international standard classification of occupation 2008 (ISCO-08) constructed with data from 11 the ISSP 2002-2007. Paper presented at the Annual Conference of International Social Survey Programme, Lisbon.

Table 9: Ten major groups of the ISCO-08 classification (single-digit)

ISCO-08 (single-digit)	Types of occupations in this group
0	Armed forces occupations
1	Managers
2	Professionals
3	Technicians and associate professionals
4	Clerical support workers
5	Service and sales workers
6	Skilled agricultural, forestry and fishery workers
7	Craft and related trades workers
8	Plant and machine operators, and assemblers
9	Elementary occupations

5. Value Scales and Personality Indicators[19]

5.1 The Inglehart Index

The VIC study is devoted to the study of values in times of crisis. Thus, it is of major relevance to conceptualize well-functioning measurements of values which enable sophisticated conclusions of value dynamics in the midst of the crisis. One of the first measurements which was integrated in the World Value Survey from the beginning, was the four-item scale of value priorities by Inglehart (1979)[20], it is called the Inglehart Index.

In the form of a ranking procedure, two materialistic and two post-materialistic goals are given, and the respondents are asked to name one goal of the first and one of second priority (Inglehart & Abramson, 1999)[21]. Based on the judgements of the respondents it is possible to assign individuals to materialists, post-materialists and mixed types. A standard variable in wave 1 as well as in wave 2 allows comparing the proportion of people being clearly materialist, favouring materialism, favouring post-materialism and falling into the category of clear postmaterialists.

5.2 The Schwartz values

In addition, VIC includes the value concept of Shalom Schwartz (1992)[22] measuring 10 individual value orientations.

Schwartz (1992)[23] has developed extensive scales trying to measure explicitly people's basic values. The ten individual value orientations form a continuum, with incompatible values opposing each other in the circular structure and values highlighting similar goals being adjacent. In addition,

[19] This section of the text is – to a large extent – identical with the same section in the methods report which was published as an attachment to the international dataset of the VIC-study, see here: https://data.aussda.at/file.xhtml?fileId=4986&version=1.2

[20] Inglehart, R. (1979). *Value priorities and socioeconomic change.* In S. Barnes & M. Kaase (Eds.), Political action: Mass participation in five Western democracies (S. 305–342). Beverly Hills: Sage Publications.

[21] Inglehart, R., & Abramson, P. R. (1999). Measuring postmaterialism. *American Political Science Review*, 93 (3), 665-677.

[22] Schwartz, S. H. (1992). Universals in the Content and Structure of Values: Theoretical Advances and Empirical Tests in 20 Countries. *Advances in Experimental Social Psychology*, 25, 1–65.

Schwartz formed two superordinate bipolar dimensions. In the dimension of openness to change vs. conservation (simply put: individualism vs. conservatism), values that push individual thinking and action or emphasize the preservation of security and tradition are opposed to each other. The second bipolar dimension of self-enhancement vs. self-transcendence (simply expressed: egoism vs. altruism) describes the striving for one's own success and dominance vs. the acceptance of others as equal.

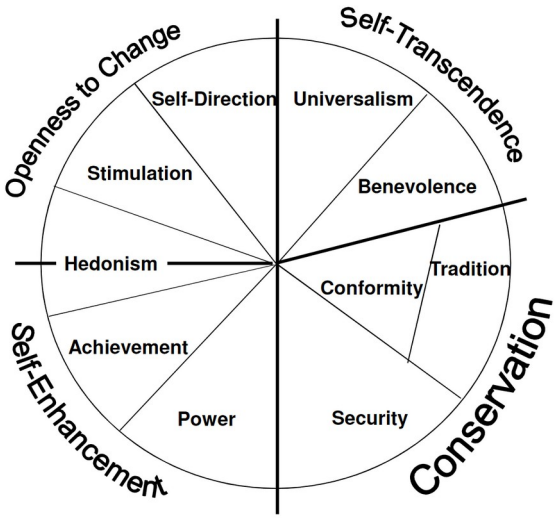

Figure 2: The classical circumplex structure of values according to Schwartz (Schwartz, 2012)[24]

After establishing his value theory Schwartz et al. (2001)[25] developed a new scale, which is more suitable for representative population surveys, called the Portrait Value Questionnaire (PVQ). This scale consists of 40 short verbal portraits of individuals. The portraits indicate the goals and expectations of the individuals and thus implicitly address the importance of their values. The respondent must indicate on a six-point scale how similar he or she feels to the person.

For the VIC study, the shortened version of the PVQ (with 22 indicators) was used, which was also implemented in the European Social Survey, one of the major survey instruments in Europe. Although this scale is also rather limited, because each value is just measured by two to three items, the scale provides a comprehensive measurement of individual value orientations. It often makes sense to calculate higher-order values and to integrate the higher-order dimensions to come to more reliable conclusions. That is the reason why the data contains different potential value measurements out of the Schwartz-scale. The ten value orientations were calculated adopting the scale use correction because it is important to correct for individual and cultural differences in scale use. We fully followed the procedures to come to reliable scales which are recommended by Schwartz, Breyer & Danner (2015)[26]. The indices of the ten individual values were used to create the

[23] Schwartz, S. H. (1992). Universals in the Content and Structure of Values: Theoretical Advances and Empirical Tests in 20 Countries. *Advances in Experimental Social Psychology*, 25, 1–65.

[24] Schwartz, S. H. (2012). An Overview of the Schwartz Theory of Basic Values. *Online Readings in Psychology and Culture*, 2(1). https://doi.org/10.9707/2307-0919.1116

[25] Schwartz, S.H., et al. (2001). Extending the cross-cultural validity of the theory of basic human values with a different method of measurement. *Journal of Cross-Cultural Psychology*, 32, 519–542.

[26] Schwartz, S. H., Breyer, B., & Danner, D. (2015). Human Values Scale (ESS). Zusammenstellung sozialwissenschaftlicher Items und Skalen (ZIS). https://doi.org/10.6102/zis234

higher order values and two bipolar dimensions of openness to change vs. conservation of order and self-enhancement vs. self-transcendence. The two opposite dimensions (which usually form a sufficient negative correlation) were subtracted from each other. All those value scores are available for VIC wave 1 and VIC wave 2.

In VIC wave 2 we additionally implemented two time points in the survey to measure how long the participants needed to complete the Schwartz value scheme. We introduced a filter variable in VIC 2 to include only participants which have at least needed more than 100 seconds to complete the scale. We advise to use this filter variable when analysing value priorities among the Austrian sample.

5.3 The measurement of personality traits

Besides common value instruments, the VIC 1 questionnaire captures personality scales. One of the well-known concepts in personality research is the BIG-Five approach[27], which states that the human personality consists of five major personality factors. In the survey, the 10-item based Big Five Inventory (BFI-10)[28] represents a short measurement of the five personality traits Extraversion, Agreeableness, Conscientiousness, Emotional Stability, and Openness. The scale was especially designed for multi-thematic questionnaires in which respondents' time is severely limited. Additionally, a few items of the sophisticated measurement of empathy by Davis (1983)[29] were used to calculate an empathy index. This computation is simply the mean of all judgements measuring empathy. Table 10 gives an overview of the measurement of the key personality traits in the survey.

Table 10: The 10 item based Big Five Inventory (BFI-10) and the empathy measurement used in the source questionnaire

Personality Traits	Items	Pole	Based on
Extraversion	I am someone who is reserved.	-	BFI 10 (Rammstedt & John, 2007), two items per dimension with five-point scale from 1 = disagree strongly to 5 = agree strongly)
	I am someone who is outgoing, sociable.	+	
Agreeableness	I am someone who is generally trusting.	+	
	I am someone who tends to find fault with others.	-	
Conscientiousness	I am someone who tends to be lazy.	-	
	I am someone who does a thorough job.	+	
Emotional lability	I am someone who is relaxed, handles stress well.	-	
	I am someone who gets nervous easily.	+	
Openness	I am someone who has few artistic interests.	-	
	I am someone who has an active imagination.	+	
Empathy	I often have tender, concerned feelings for people less fortunate than me.	+	Based on Davis (1983) (selection of 4 out of 28 items, five-point scale from 1 = does not describe me at all to 5 = describes me perfectly well
	When I see people being taken advantage of, I feel kind of protective towards them.	+	
	I am often quite touched by things that I see happen.	+	
	I would describe myself as a pretty soft-hearted person.	+	

[27] Costa Jr, P. T., & McCrae, R. R. (1992). The five-factor model of personality and its relevance to personality disorders. *Journal of personality disorders*, 6(4), 343-359.

[28] Rammstedt, B., & John, O. P. (2007). Measuring personality in one minute or less: A 10-item short version of the Big Five Inventory in English and German. *Journal of research in Personality*, 41 (1), 203-212.

[29] Davis, M. H. (1983). Measuring individual differences in empathy: Evidence for a multidimensional approach. *Journal of personality and social psychology*, 44(1), 113-126.

6. Data quality assessment

To assess data quality, we conducted multiple checks involving age and sex information. Two data sources were utilized for this evaluation: the registration data supplied by online access panel members during their sign-up, and information gathered independently across all three survey waves. In total, we conducted twelve distinct tests. In Figure 1, however, only 11 tests are listed instead of 12. Apparently, the frequency data for sex W1W2 are missing. We compared responses for age and sex separately across different wave pairs (W1W2, W2W3, W1W3) and between individual waves (W1, W2, W3) and the registration data. Any disparities between these comparison points were recorded as anomalies. In the case of sex, discrepancies were evident based on predefined categories. Concerning age, anomalies were identified as any implausible age changes, such as someone becoming younger or aging faster than the time intervals between waves would allow. This process is inherently imprecise for age, as even a minor age increase could potentially be flagged as an anomaly. Consequently, the age test is a more conservative estimate compared to the sex test.

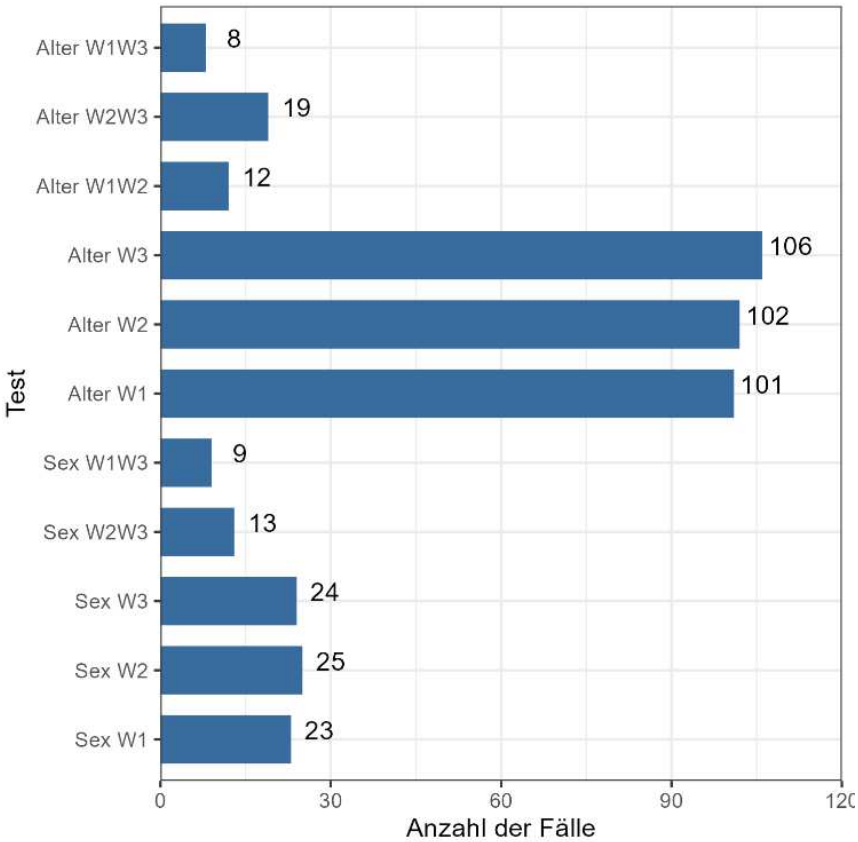

Figure 3: Frequencies of anomalies in the 12 tests

Out of the total 3,797 cases that participated in at least one of the three waves, 3,490 (91.9%) displayed no anomalies. The remaining cases are distributed as shown in Table 11.

Table 11: Sum of anomalies across 12 tests by cases (n=3797)

Sum of anomalies	Cases	Percentage
0	3490	91.91
1	228	6.00
2	46	1.21
3	16	0.42
4	13	0.34
5	2	0.05
6	2	0.05

The dataset includes a variable (Issue_Sum) that provides the sum of anomalies for each record.